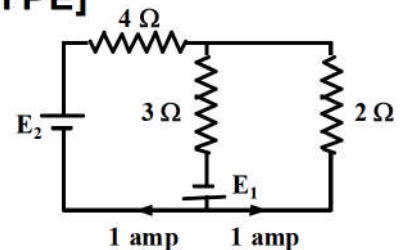


WORK SHEET - 22

CURRENT ELECTRICITY

[SINGLE CORRECT CHOICE TYPE]

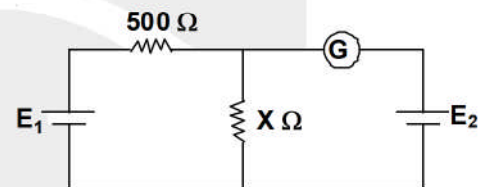
- Q.1 In the adjoining circuit diagram, the emf of the cells E_1 and E_2 are respectively : [3]
(A) 2 V, 9 V (B) 8 V, 2 V
(C) 11 V, 2 V (D) 11 V, 9 V



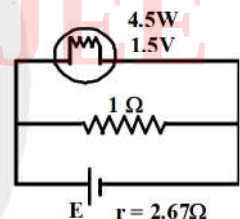
- Q.2 A battery of 10 volt is connected to a resistance of 20 ohm through a variable resistance R . Find the amount of charge which has passed in the circuit in 4 minutes, if the variable resistance R is increased at the rate of 5 ohm/min starting from zero. [3]

- (A) 120 C (B) $120 \log_e 2$ C (C) $\frac{120}{\log_e 2}$ C (D) $\frac{60}{\log_e 2}$ C

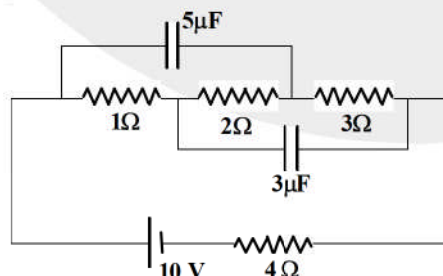
- Q.3 In the adjoining circuit, the battery E_1 has an E.M.F. of 12 volts and zero internal resistance. While the battery E_2 has an E.M.F. of 2 volts. If the galvanometer G reads zero then the value of the resistance X in ohms is [3]
(A) 10 (B) 100
(C) 14 (D) 200



- Q.4 A bulb rated as 4.5 W, 1.5V is connected as shown. The emf of the cell, needed to make the bulb glow at full intensity is [3]
(A) 4.5 V (B) 1.5 V (C) 2.67 V (D) 13.5 V

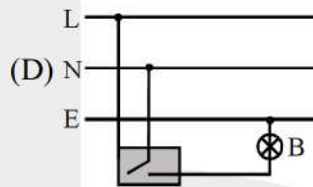
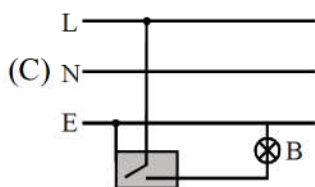
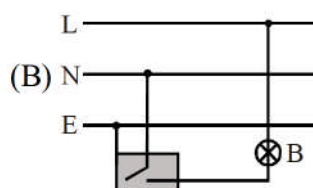
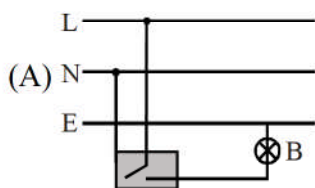


- Q.5 An electric current of 16 A exists in a metal wire of cross section 10^{-6} m^2 and length 1m. Assuming one free electrons per atom. The drift speed of the free electrons in the wire will be [3]
(Density of metal $= 5 \times 10^3 \text{ kg/m}^3$, atomic weight = 60)
(A) $5 \times 10^{-3} \text{ m/s}$ (B) $2 \times 10^{-3} \text{ m/s}$ (C) $4 \times 10^{-3} \text{ m/s}$ (D) $7.5 \times 10^{-3} \text{ m/s}$
- Q.6 In the shown network, charges on capacitors of capacitances $5 \mu\text{F}$, $3 \mu\text{F}$ in steady state will be [3]



- (A) $15 \mu\text{C}$, $15 \mu\text{C}$, (B) $15 \mu\text{C}$, $5 \mu\text{C}$, (C) $5 \mu\text{C}$, $15 \mu\text{C}$, (D) $5 \mu\text{C}$, $5 \mu\text{C}$,
- Q.7 If a cell of constant emf produces the same amount of heat during the same time in two independent resistors 9Ω and 4Ω , when they are separately connected across the terminals of the cell, one after another, the internal resistance of the cell is [3]
(A) 2.5Ω (B) 6Ω (C) 5Ω (D) 3Ω

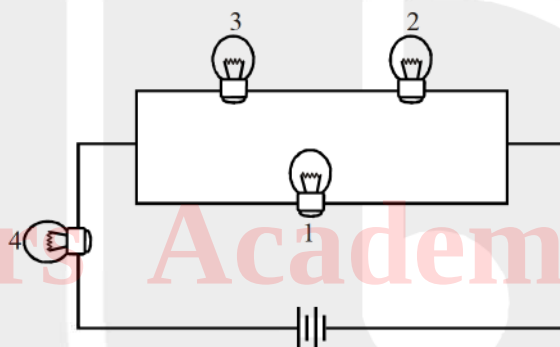
- Q.8 The circuit showing the correct connections for a switch and lamp in a lightening circuit is [L → Live ; N → Neutral ; E → Earth ;  → metal case] [3]



[PARAGRAPH TYPE]

Paragraph for question nos. 9 to 11

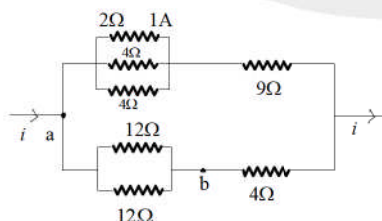
All bulbs consume same power. The resistance of bulb 1 is 36Ω .



- Q.9 What is the resistance of bulb 3 : [3]
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
- Q.10 What is the resistance of bulb 4 : [3]
 (A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω
- Q.11 What is the voltage output of the battery if the power of each bulb is $4W$: [3]
 (A) $12V$ (B) $16V$ (C) $24V$ (D) None of these

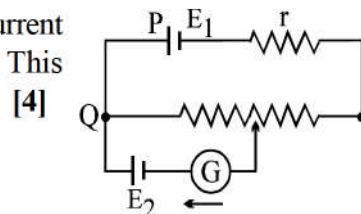
[MULTIPLE CORRECT CHOICE TYPE]

- Q.12 Consider the network shown in the figure. If the current in 2Ω resistor is $1A$, then [4]



- (A) potential difference across points a & b equals $6V$.
 (B) current through all 4Ω resistors equals $2A$.
 (C) total power developed in the 12Ω resistors equals 24 watts.
 (D) none of the above

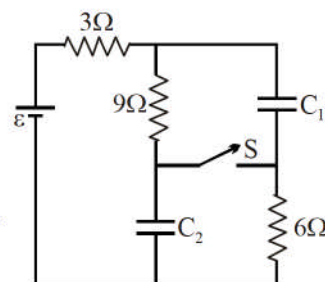
- Q.13 In the potentiometer circuit of given figure the galvanometer reveals a current in the direction shown wherever the sliding contact touches the wire. This could be caused by
 (A) E_1 being too low
 (B) r being too high
 (C) a break in PQ
 (D) E_2 being too low



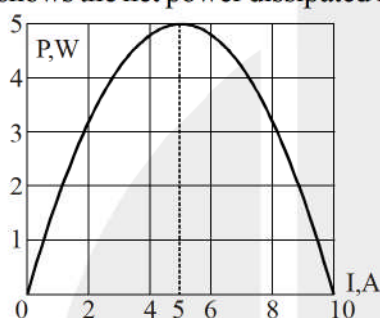
- Q.14 In the circuit shown there is steady state with the switch closed. The switch is opened at $t = 0$. Choose the correct option(s).

(Given : $\varepsilon = 24\text{V}$, $C_1 = 3\text{F}$ and $C_2 = 2\text{F}$)

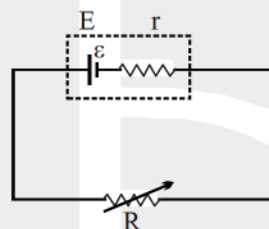
- (A) The voltage across C_1 before the switch is open is 12V.
 (B) The voltage across C_1 after a long time after the switch is open is 12V.
 (C) The voltage across C_2 after a long time after the switch is open is 24V.
 (D) The voltage across C_2 before the switch is open is 8V.



- Q.15 Figure shows the net power dissipated in R versus the current in a simple circuit shown.



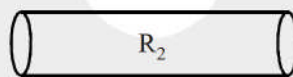
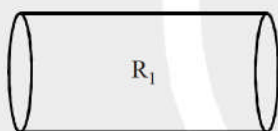
- (A) The internal resistance of battery is 0.2Ω .
 (C) R at which power is 5W is 2.5Ω



- (B) The emf of battery is 2V
 (D) at $i = 2\text{A}$, power is 3.2 W

[MATRIX TYPE]

- Q.16 Figure shows two cylindrical uniform specimen of electrical resistances with their characteristics. [8]



Resistivity :

Length :

Area of cross-section :

No. of electron per unit volume :

$$\rho_1 = \rho$$

$$\ell_1 = \ell$$

$$A_1 = A$$

$$n_1 = n$$

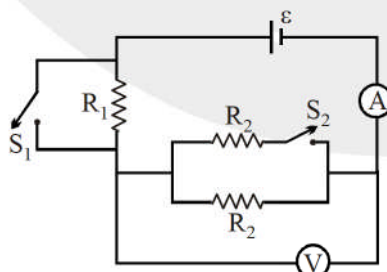
$$\rho_2 = 2\rho$$

$$\ell_2 = 2\ell$$

$$A_2 = A/2$$

$$n_2 = 2n$$

R_1 and R_2 are connected in circuit as shown in figure. Voltmeter and ammeter connected in the circuit are ideal. Comparison of electric field, drift speed and power dissipated per unit volume should be done only if R_1 carries current.



Column-I

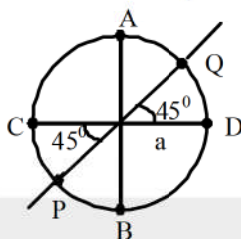
- (A) S_1 is open S_2 is closed
 (B) S_1 is closed S_2 is open
 (C) S_1 and S_2 both are closed
 (D) S_1 and S_2 both are open

Column-II

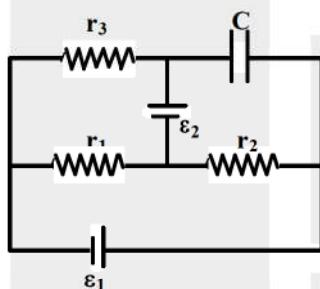
- (P) Reading of voltmeter is maximum
 (Q) Reading of ammeter is maximum
 (R) Magnitude electric field in R_2 is greater than in R_1 .
 (S) Drift speed of electrons is greater in R_2 than in R_1 .
 (T) Power dissipated per unit volume is greater in R_2 than in R_1 .

[SUBJECTIVE TYPE]

- Q.17 Consider a wire loop as shown in the figure. If the resistance per unit length is λ , find the effective resistance if battery is connected between P & Q. [5]

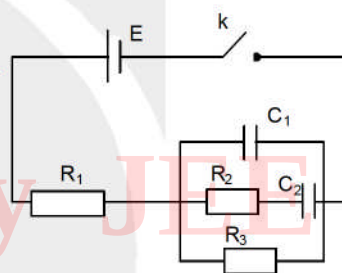


- Q.18 In steady state condition charge on the capacitor is _____. $E_1 = 2$ volt, $E_2 = 4$ volt $r_1 = 1\Omega$, $r_2 = 1\Omega$, $r_3 = 2\Omega$, $C = 1\mu F$ [5]

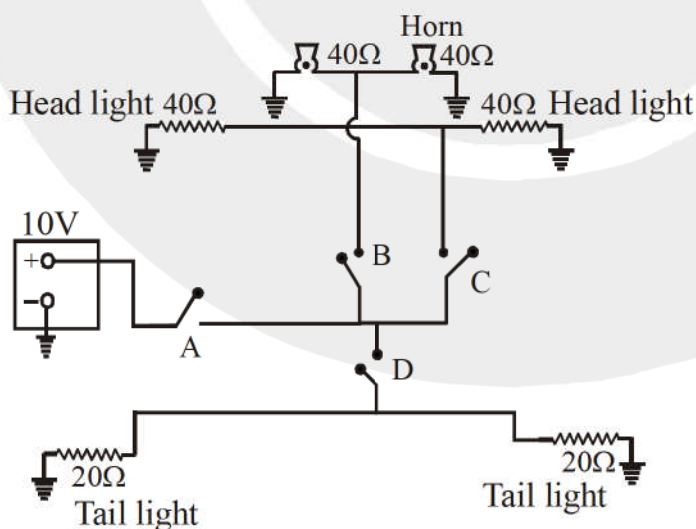


- Q.19 Determine the current through the battery in the circuit shown in the figure. [5]

- (i) Immediately after the key k is closed.
(ii) After long time key k is closed.



- Q.20 Figure shows an automobile circuit. How much power (in watt) is dissipated by the automobile circuit when switches A, B, C and D are all closed. [5]



ANSWER KEY

Q.1	B	Q.2	B	Q.3	B	Q.4	D	Q.5	B	Q.6	A	Q.7	B
Q.8	B	Q.9	B	Q.10	A	Q.11	B, C	Q.13	A, B, C		Q.14	A, C, D	
Q.15	A, B, D	Q.16	(A) R, T (B) P (C) P, Q (D) R, T	Q.17	$\frac{2\pi a\lambda(8+\pi)}{32+\pi^2+16\pi}$								
Q.18	$2\mu C$	Q.19	(i) E/R_1 (ii) $E/(R_1+R_3)$	Q.20	0020								

Rankers Academy JEE

